Abbas Taqvi

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Personal Information

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Current Location	Milan, Italy
Languages	English (CEFR C1) Italian (CEFR A2) Urdu (native)

Professional Summary

Data engineer with a founding-engineer background, currently pursuing an M.S. in Data Science at Milano-Bicocca. Built a lakehouse platform that queries 30B+ records in under a second, and a Rust ETL framework that runs 10× faster than the system it replaced. Seeking a data engineering internship in Italy.

Technical Skills

- **Programming Languages:** Python, Rust (incl. embedded Rust, tokio), Scala, C, C++ (CUDA), Go, SQL, NoSQL query languages, JavaScript, Dart, R, Verilog, RISC-V Assembly
- **Data Engineering:** Apache Spark (Scala, PySpark), Apache Kafka, Apache Airflow (dynamic DAGs), Trino / Presto / Athena, Apache Iceberg, Hive / Hadoop, Databricks, Apache NiFi, Informatica, AWS S3, AWS Glue, AWS Lake Formation, lakehouse / data lake / data mesh / data warehouse architecture, ETL / ELT, streaming and batch ingestion, medallion (bronze, silver, gold) layers, JSON and Parquet processing, schema management, file compaction and partitioning
- **AI / ML Engineering:** PyTorch, TensorFlow, Keras, JAX, NumPyro, LangChain, LangGraph, Pydantic-AI, RAG, Text2SQL and multi-agent systems, Gemini API, Hugging Face transformers (BERT), Word2Vec, normalizing flows, diffusion models, CUDA / GPU programming
- **Databases and Vector Stores:** PostgreSQL, MongoDB, Neo4j, Amazon RDS, Qdrant, Weaviate
- **Cloud, DevOps and Infrastructure:** AWS (S3, EKS, EC2, Glue, Lake Formation, RDS), Azure (Databricks), GCP (Cloud Functions, GCS, BigQuery), Docker, Kubernetes (EKS, MicroK8s), Helm, Traefik, Cloudflare (Tunnels, Pages), Linux (Ubuntu, long-time user), macOS, Git, CI/CD, self-hosting
- **Frameworks and Tools:** FastAPI, Django, Django REST Framework, Flutter, React, Scrapy, Playwright, Budibase, Power BI, DataGrip, Postman, Pandas, NumPy, Matplotlib, R Markdown
- **Engineering Practices:** Architecture planning, system design, design patterns, REST APIs, gRPC, root cause analysis, debugging, benchmarking and performance optimization

Work Experience

Data Engineer I (Founding Engineer) — United States (Remote)
Logiciel Services LLC — May 2024 – Nov 2025

- Recruited as a Founding Engineer in the Data and AI department; designed and built the company's entire

backend data platform end-to-end: Kafka streaming ingestion, Spark (Scala) processing, S3 lakehouse storage, Trino query engine and Airflow orchestration. Simplified a 12-component architecture into 5.

- Set up data lake, data mesh and data warehouse layers on S3 with Trino for federated queries and AWS Glue for cataloguing, housing financial data of 30B+ records that can be queried in under a second with appropriate filters.
- Developed Spark pipelines in Scala for ingestion and transformation of US stock market FIX logs: 375+ FIX sessions daily across multiple products and environments, with streaming and batch ingestion, orchestrated through Airflow dynamic DAGs auto-generated from YAML configuration.
- Improved operational reliability with schema control, file compaction logic and structured partitioning.
- Wrote Ferroflow, a Rust ETL framework with composable processor and pipeline abstractions for end-of-day processing of FIX logs, running alongside a NiFi pipeline and replacing a legacy .NET utility: 500 min $\rightarrow$ 50 min daily (10$\times$ speedup, 90% runtime reduction), saving 400 EC2-minutes per run.
- Built Metosi, a domain-specific multi-agent Text2SQL chatbot (Python, LangGraph, Trino) that lets business teams and stakeholders at Velocity Capital query the data lake in natural language through a REST API, reducing analyst turnaround by 41%.

DevOps Intern — San Jose, CA (Remote)
Pullflow Inc. — Jul 2023 – Aug 2023

- Migrated production infrastructure from AWS EKS to on-premise Kubernetes (MicroK8s) and Docker: wrote Dockerfiles, Helm charts and a Traefik reverse-proxy configuration for load balancing.
- Deployed a self-hosted Budibase analytics dashboard on a custom subdomain with a zero-downtime cutover.

Data Engineering Intern — Karachi, Pakistan (On-site)
United Bank Limited — May 2023 – Jun 2023

- Built a cross-region streaming pipeline in Informatica that ingested data from the Qatar and Bahrain regions into a Hive data lake.
- Took part in architecture analysis and validated production AI pipelines against new regional schemas across 3 countries (Pakistan, Qatar, Bahrain).

Junior Data Engineer — Pinedale, Wyoming, US
ImproData — (Remote)
Jan 2021 – May 2022

- Built risk-management data pipelines for cyber-security data on Azure Databricks (PySpark), structured as bronze, silver and gold layers.
- Automated bulk ingestion with GCP Cloud Functions, moving compressed JSON from local storage to GCS and loading it into BigQuery (JSON $\rightarrow$ GCS $\rightarrow$ BigQuery), eliminating about 3 hours of daily manual work.

Data Researcher — Habib University, Karachi
Office of Institutional Research — Aug 2021 – Sep 2021

- Collected, cleaned and manipulated institutional data with NumPy, Pandas and Excel to support research and reporting.

Data Analysis Intern — Remote (Volunteer)
EdLab Pakistan — Sep 2021 – Oct 2021

- Performed exploratory data analysis and built visualizations of key trends with Power BI and Excel.

Teaching Assistant — Habib University, Karachi
Computer Science courses — Spring 2023 – Spring 2024

- **Data Structures and Algorithms (Spring 2023):** assisted students in labs and with assignments.
- **Object-Oriented Programming and Design Methodologies (Fall 2023):** guided students through design patterns and object-oriented programming concepts.
- **Algorithm Design and Analysis (Spring 2024):** helped students build a foundation in algorithm analysis and computational complexity.

Workshop Instructor — Habib University, Karachi
Git and GitHub — 2023

- Conducted two workshops on the fundamentals of Git and GitHub, covering version control in collaborative software development.

Education and Training

Università degli Studi di Milano-Bicocca Milan, Italy
M.S. in Data Science Oct 2025 – Present
- Coursework includes Data Semantics, Technological Infrastructures for Data Science and Statistical Modeling.

Habib University Karachi, Pakistan
B.S. in Computer Science Aug 2020 – May 2024
- CGPA 3.31 (out of 4.0); HU-TOPS 100% merit scholarship for the full four-year program.
- Final year project: Astroflow, Bayesian inference with normalizing flows (see Projects and Research).

Projects

Industry projects

Kuza: End-to-End FIX Data Platform Logiciel Services LLC
Scala, Spark, Kafka, Airflow, Trino, AWS S3, AWS Glue, PostgreSQL, Docker Jan 2025 – Nov 2025
- Implemented FIX ingestion pipelines for 375+ daily sessions across multiple environments, with Spark (Scala) workflows orchestrated by Airflow dynamic DAGs auto-generated from YAML.
- Built an S3-based data lake with Trino and AWS Glue for unified schema management and fast querying; streamed events through Kafka.
- Improved reliability with schema control, file compaction logic and structured partitioning; reduced a fragmented 12-service topology to a 5-component architecture.

Ferroflow: Rust ETL Framework Logiciel Services LLC
Rust Jun 2024 – Jul 2024
- Architected a Rust ETL framework inspired by NiFi, with reusable processors, composable pipelines and zero-cost ownership semantics.
- Achieved a 10× performance gain, cutting FIX-log processing from 500 minutes to about 50 minutes and saving 400 EC2-minutes per run.
- Implemented optimized chunking, streaming and memory-mapped file strategies for large-file processing.
- Integrated Ferroflow in hybrid mode with NiFi to allow a smooth migration away from legacy .NET workflows.

Metosi: FIX Logs Multi-Agent Chatbot Logiciel Services LLC
Python, LangChain, LangGraph, SQL, Trino, REST Sep 2024 – Jan 2025
- Built a multi-agent chatbot grounded in FIX logs, trade events and FinTech operational concepts, exposed through a REST API.
- Connected the chatbot to Trino for real-time SQL queries over a 30B-record lake, enabling direct analytics without writing SQL.
- Added retrieval, reasoning and execution flows with LangGraph for deterministic and interpretable outputs.
- Reduced manual analyst effort by about 41%.

Academic and research projects

Astroflow: Bayesian Flow Networks (Final Year Project) Habib University
PyTorch, Django, PostgreSQL Aug 2023 – May 2024
- Developed a Bayesian inference framework using normalizing flows for binary black-hole population modeling, under the guidance of Muhammad Zeeshan (PhD candidate, Rochester Institute of Technology).
- Designed neural summarizers and feature extractors for unseen gravitational-wave-like data.
- Reduced parameter inference from roughly 300 minutes with traditional statistical methods to about 2 seconds per inference.
- Optimized PyTorch training loops for faster posterior sampling and better model stability; built a Django interface to run inference pipelines, organize experiments and visualize posterior samples.

Course research projects Habib University
3rd–8th semester
- **Pedometer on RISC-V (3rd semester):** designed and implemented a pedometer system on the RISC-V architecture under Dr. Owais Waheed Talaat.

- **CUDA Path Tracer with Binary Interval Hierarchies (6th semester):** GPU-accelerated path tracer in CUDA, focused on efficient data flow and computational balance.
- **Applications of LLMs in Software Engineering (7th semester):** investigated downstream tasks of large language models, in particular automated code generation and maintenance.
- **Explainable AI with Diffusion Models (8th semester):** explored explainability techniques using diffusion models in a computer vision course.
- **Rust-based Compiler Construction:** research on compiler construction in Rust with Dr. Taj Muhammad Khan, focused on low-level optimizations in the compilation process.

Personal and hackathon projects

Yohsin: High-Performance Serialization Tool Hackathon
Rust, tokio, mmap Jan 2025

- Built, in 24 hours, a zero-copy, multi-threaded serializer and deserializer that sustains about 1M records (2.25 GB) per second.
- Used memory-mapped files, offset-indexed headers and async I/O for O(1) random access and low RAM overhead; added range deserialization and state checkpointing.
- Added fault tolerance through an append-only write-ahead log with stateful recovery in under 50 ms, and built benchmark suites across record sizes and workloads.

Bayaaz: Mobile App Live on Google Play
Flutter, FastAPI, PostgreSQL, Cloudflare, React and App Store

- Published a Flutter poetry app on both Google Play and the App Store, and maintain its backend: a stateless FastAPI service on PostgreSQL, exposed through Cloudflare tunnels, with a React frontend on Cloudflare Pages.
- Evaluated self-hosted and cloud hosting options for cost and reliability at small scale, and over-the-air update options for Flutter with CI/CD discipline.

Research and Conferences

- **Poster, The Computing Conference 2025 (SAI Conferences), London, UK:** "Astroflow: Accelerating Parameter Inference of Binary Black Holes' Population using Flow-based Neural Networks." The work uses flow-based neural networks to reduce inference time from about 300 minutes with traditional statistical methods to about 2 seconds, and was supervised by Muhammad Zeeshan (PhD candidate, Rochester Institute of Technology).

Achievements

- **HU-TOPS 100% Scholarship** Habib University
  Merit scholarship for the B.S. in Computer Science 2020 – 2024
- **Winner, Game of Codes 5.0** Green Island Youth Foundation
  Competitive programming competition 2023
- **6th position country-wide, Google Kickstart (May 2022)** Google
  Competitive programming competition 2022
- **19th position, ICPC Asia Topi Regionals 2022** ICPC
  International Collegiate Programming Contest 2022

Test Scores

- **IELTS (International English Language Testing System):** 8.0 bands (out of 9.0)
- **GRE (Graduate Record Examinations), Quantitative:** 165/170